PARMCO – AI Interaction Log (Claude)

ECSE 4235 · Group 2 · Checkpoint 3 work session

Project	PARMCO – Phone App RP4 Motor Controller
Team	Elvin Alvarez, Maya Nessmith
AI tool	Claude (Anthropic), Claude Opus 5.5, via the Claude desktop app with access to the team's project folder
Session dates	Nov 10, 2026, 4:21 PM – Nov 11, 2026, 3:37 PM (Eastern)
Prepared by	Claude, at the team's request

About this log

- Every prompt from the team is reproduced word for word, including pasted console output.
- Claude's responses are condensed to what was done, what was decided and which files were created or changed, so the work can be replicated without reproducing every line of every reply. The complete raw transcript can also be exported from claude.ai (Settings → Privacy → Export data).
- All code, circuit drawings and website files described here were authored by Claude. Human edits made by the team are noted where Claude observed them; the team marked them in the files with [HUMAN EDIT] … [END HUMAN EDIT].
- Times are when the prompt was sent (US Eastern).

Session overview

#	Phase	Exchanges	What happened
1	Requirements and audit	1–3	Broke down the project document; audited CP1/CP2/CP3 code, class library and documentation against the requirements.
2	Hardware and schematic	4–10	Power-delivery review; LTspice-style schematic (Rev A → Rev B), colour-coded wiring.
3	CP3 software and iPhone app	11–15	Reviewed partner's CP3 code; built BLE server, rewrote the RP4 controller, assembly step-timing loop and SwiftUI iPhone app; I²C questions.
4	Bench debugging	16–36	Bluetooth pairing loop, encoder direction, angle-target accuracy, CW/CCW direction, Test B encoder readings, version control problems.
5	GitHub site	37–45, 48–52, 54–56	Site requirements; skeleton; full rebuild with UGA theme and scroll-driven propeller; GitHub Pages setup and deployment.
6	Schematic Rev C and app preview	46–47, 53	Added AS5600 encoder and common-ground network; Xcode preview/demo mode; link styling.

#	Phase	Exchanges	What happened
7	Export	57	This log.

Exchange log

1 Nov 10, 4:21 PM

Prompt (team, verbatim)

> [Attached: ECSE4235_PARMCO_Project.pdf] I am recruiting your help with a class project designed to give us experience in using AI tools in engineering. Attached is the documentation, highlighting all of the constraints and requirements for the project. Additionally, some work has already been completed; up to checkpoint 2 has been completed. however, we will need to retroactively work on some of the previous goals already met. I will shortly attach any other required material for you to understand the project and map out where we are at, in terms of progress and meeting the project requirements. Attached to this chat is the project intructional documentation. I need you to break down each requirement and objective first before we continue.

Response (Claude, condensed)

- Read the full 45-page project document.
- Broke down: the system, AI-authorship rules, hard technical rules (bench supplies, 4N25 enable with no PWM, ARM assembly H-bridge interface, class library for all GPIO), every phone-app function, Bluetooth, logging specification, keyboard commands, Tests A–G, and each checkpoint's deliverables and grading.
- Listed easy-to-miss requirements (reject not clamp out-of-range RPM, unlock auto-off on stop, refuse set-to-zero before calibration, coil re-alignment in the feedback hold, monotonic-clock logging, auto-start).
- Flagged that CP3 was two days away.

Files: —

2 Nov 10, 10:44 PM

Prompt (team, verbatim)

> [Attached: ECSE_4235_Library_Documentation.pdf; project folder connected] attached is all of our group's code and some documentation for the first 2 checkpoints. additionally, i have attached the documentation for the class library

Response (Claude, condensed)

- Read the class library documentation (GPIO, I²C, keyboard functions; no interrupts or timers).
- Audited CP1/CP2 code, App Inventor project, schematics and AI logs.
- Compliance findings: the IR polling read GPIO registers directly (gpio_mem_ptr) instead of E4235_Read; the step-timing loop did not call the step routine and drifted; no human-edit markers; several CP1/CP2 documents not found in the folder; no per-checkpoint repos.
- Listed what CP3 still needed (merged program, encoder, calibrate/set-zero/angle, telemetry, logging, unlock, app changes) and proposed a 36-hour plan.

Files: —

3 Nov 10, 10:48 PM

Prompt (team, verbatim)

> appologies, i forgot to pull from git the most recent updates. can you recheck the folder?

Response (Claude, condensed)

- Reviewed the partner's new CP3/parmco_cp3_controller.c.
- Credited: merged threads, start/stop/direction/faster/slower, RPM validation, defaults, logging start.
- Gaps scored against the CP3 peer check (~25/60 points): no encoder support, IR RPM quantised by a 1-second count window, no unlock, minimal telemetry, direct register read, log format, no defaults on connect.

Files: —

4 Nov 10, 11:09 PM

Prompt (team, verbatim)

> my lab mate is currently working on app updates, as we speak, so we will focus on something a little different for the time being. The motor struggles to reach the minimum requirement of 100 rpm at times. I want us to look over the wiring and schematic for the motor circuit to ensure power delivery is not an issue.

Response (Claude, condensed)

- Checked wiring/netlist against SN754410 and motor datasheets: wiring correct.
- Explained the built-in limits: ~2.1 V driver drop (≈0.27 A instead of 0.35 A per phase) and coil L/R time constant near 100 RPM.
- Listed intermittent suspects (supply current limit, driver heating, no bulk capacitor, Linux step jitter, no start ramp) with a 15-minute bench test plan and ranked fixes.

Files: —

5 Nov 10, 11:21 PM

Prompt (team, verbatim)

> can you give me an image of the schematic, similar to one you would see from LTSpice?

Response (Claude, condensed)

- Wrote a Python generator producing an LTspice-style schematic (U1 RP4, U2 4N25, U3 SN754410NE, U4 IR sensor, M1 motor, R1–R6, C1–C2, bench supplies, net labels, notes).
- Proposed bulk capacitors C3/C4 drawn as red dashed revisions.

Files: parmco_motor_schematic.png / .svg (Rev A)

6 Nov 11, 12:55 AM

Prompt (team, verbatim)

> the motor now spins much better and schematic is verified as correct

Response (Claude, condensed)

- Recommended re-running the stall characterisation (3 runs) and asked what changed so the records stay accurate.

Files: —

7 Nov 11, 1:02 AM

Prompt (team, verbatim)

> no, we kept the 100nF capacitors at the 5 and 12 v rails. it is a possibility the coils were mismatched. I had taken the whole circuit apart prior to verifying to rebuild it from scratch

Response (Claude, condensed)

- Agreed miswired coil pairs explain the erratic CP1 stall data.
- Updated schematic to Rev B: removed proposed C3/C4, added a revision note (rebuild, coil pairs re-verified).

Files: parmco_motor_schematic.png / .svg (Rev B)

8 Nov 11, 1:05 AM

Prompt (team, verbatim)

> could you make some minor changes to the schematic. could you use green lines for power lines, yellow, for signals coming out of the pi, orange for signals going between IC components and black for grounds? and of course, color coordinate the motor connects to reflect real life as well

Response (Claude, condensed)

- Recoloured wires by type, coloured motor leads by real lead colour, added a colour key.

Files: parmco_motor_schematic.png / .svg

9 Nov 11, 1:08 AM

Prompt (team, verbatim)

> almost. IR_OUT needs to be yellow. could you also make the motor yellow wire visibly darker than pi signal nodes so they are easier to distinguish

Response (Claude, condensed)

- IR_OUT changed to yellow; motor YEL lead darkened to gold.

Files: parmco_motor_schematic.png / .svg

10 Nov 11, 1:09 AM

Prompt (team, verbatim)

> same with the green motor lead please

Response (Claude, condensed)

- Motor GRN lead darkened to distinguish it from power lines.

Files: parmco_motor_schematic.png / .svg

11 Nov 11, 1:34 AM

Prompt (team, verbatim)

> you can check the final project folder again. there were some updates pushed onto the git repo

Response (Claude, condensed)

- Reviewed new AS5600 module and controller changes.
- Gaps: SET_ZERO only re-zeroed the encoder (did not move), angle target not implemented, no calibration state, noisy 10 ms encoder RPM, 1 s IR window, no unlock, direct register read, Linux i2c-dev vs class library, log format.
- Noted the Kotlin app source was not in the repo.

Files: —

12 Nov 11, 1:46 AM

Prompt (team, verbatim)

> alright all files are confirmed up to date now. check you have the most updated files

Response (Claude, condensed)

- Confirmed all files current; noted the new log only had a header row; asked for encoder direction and app command protocol.

Files: —

13 Nov 11, 1:52 AM

Prompt (team, verbatim)

> i dont have the android device with me. we should go ahead and create a new app for me to be able to test and confirm meets the checkpoint 3 criteria. I have an iphone 16 pro, running iOS 27.0.1. I have XCode installed on my mac with iOS 27.2 simulator.

Response (Claude, condensed)

- Explained iPhones cannot use Bluetooth Classic serial (RFCOMM) and the Simulator has no Bluetooth; asked the team to choose. Team chose: iOS app + BLE on the Pi; iPhone as the CP3 demo phone.
- Added ARM assembly step_at_deadline (absolute-deadline step-timing loop that calls step_routine) and step-index get/set.
- Wrote ble_server.c (BlueZ D-Bus GATT server, Nordic-UART-style service, chunked notifications) alongside the existing RFCOMM server.
- Rewrote the RP4 controller: shared command parser, encoder-driven calibrate/set-zero/angle target with calibration state, windowed encoder RPM, IR RPM from pulse period via E4235_Read, unlock (auto-off on stop), ramp, real-time motor thread, 100 Hz logging with the specified columns, 5 Hz telemetry, defaults on connect.
- Built a SwiftUI iPhone app (CoreBluetooth, remembered peripheral, all CP3 controls, validation, logging indicator, unlock) and Xcode project; README with protocol and library attribution.
- Verified with a desktop simulation of motor/encoder/IR and a mock D-Bus test.

Files: CP3/Code: parmco_cp3_controller.c, ble_server.c/.h, motor_driver.s, Makefile(.new), README.md; CP3/iOS/PARMCO (Xcode project, PARMCOApp.swift, PARMCOLink.swift, ContentView.swift)

14 Nov 11, 2:43 AM

Prompt (team, verbatim)

> what pins did you assign for the encoder?

Response (Claude, condensed)

- AS5600 on hardware I²C-1: SDA GPIO2 (pin 3), SCL GPIO3 (pin 5), 3.3 V, GND, DIR to GND; how to enable I²C and check with i2cdetect (address 0x36).

Files: —

15 Nov 11, 2:59 AM

Prompt (team, verbatim)

> will i have to reenable i2c after a power cycle?

Response (Claude, condensed)

- No – raspi-config setting persists; verify with ls /dev/i2c-1.

Files: —

16 Nov 11, 3:01 AM

Prompt (team, verbatim)

> bluetooth continues to disconnect and reconnect. after each disconnect and connect cycle, it reprompts the confirm pair request pop up as well
>
> [Pi console output pasted, showing:] group-2@ECSE4235-2:~/Documents/Group-2/finalProj/CP3/Code $ sudo ./parmco_cp3 / AS5600 encoder ready / PARMCO CP3 controller: step mode FULL (200 steps/rev), timer 54000000 Hz / Defaults: Manual, CW, 20 RPM, stopped, not calibrated / RFCOMM (Android) listening on channel 1 / [BLE] advertising as "PARMCO" / [BLE] GATT application registered / [BLE] LE advertisement registered / BLE phone connected: defaults applied / <- HELLO … <- CALIBRATE_ZERO -> ACK:CALIBRATE_ZERO / <- START -> ACK:START / [BLE] phone disconnected / BLE phone disconnected: motor stopped / … (cycle repeats)

Response (Claude, condensed)

- Diagnosed a failing pairing (no agent on the Pi) plus possible CPU starvation from the real-time motor thread.
- Added a NoInputNoOutput pairing agent that accepts and trusts the phone; changed the motor thread to sleep until ~200 µs before each step, pin to core 3, lower priority.

Files: CP3/Code/ble_server.c, parmco_cp3_controller.c

17 Nov 11, 3:08 AM

Prompt (team, verbatim)

> the bluetooth problem is fixed. however the calibration does not work. i do not get the encoder not detected error. it just doest not calibrate [interrupted by user]

Response (Claude, condensed)

- (No response – interrupted.)

Files: —

18 Nov 11, 3:24 AM

Prompt (team, verbatim)

> ignore that last chat. the error is produced because the motor is not mounted rigidly onto the desk

Response (Claude, condensed)

- Explained reaction torque shifting the stand; advised clamping, re-checking calibration each session, and recording the finding for the writeup.

Files: —

19 Nov 11, 3:24 AM

Prompt (team, verbatim)

> ls

Response (Claude, condensed)

- Listed the CP3 folder; suggested removing the stale binary and old logs and ignoring Xcode user data.

Files: —

20 Nov 11, 3:34 AM

Prompt (team, verbatim)

> i got the disconnect and reconnect bug again

Response (Claude, condensed)

- Asked for btmon disconnect reasons, timing, power (vcgencmd get_throttled) and Wi-Fi checks.

Files: —

21 Nov 11, 3:35 AM

Prompt (team, verbatim)

> SMP: Pairing Public Key (0x0c) len 64 / SMP: Pairing Public Key (0x0c) len 64 / SMP: Pairing Confirm (0x03) len 16 / SMP: Pairing Random (0x04) len 16 / SMP: Pairing Random (0x04) len 16 / SMP: Pairing Failed (0x05) len 1 / Reason: Numeric comparison failed (0x0c) / SMP: Pairing DHKey Check (0x0d) len 16 / SMP: Pairing Request (0x01) len 6 / SMP: Pairing Response (0x02) len 6 / … (sequence repeats with the same failure)

Response (Claude, condensed)

- Identified another agent (desktop Bluetooth applet) answering pairing with a confirm dialog nobody accepted.
- Made the program re-claim the default agent whenever a phone connects.

Files: CP3/Code/ble_server.c

22 Nov 11, 3:37 AM

Prompt (team, verbatim)

> can you update the file again?

Response (Claude, condensed)

- Found the file had been reverted on the Mac; rewrote and verified it; advised committing before pulling.

Files: CP3/Code/ble_server.c

23 Nov 11, 3:42 AM

Prompt (team, verbatim)

> it connected but now it is not moving onto the controller

Response (Claude, condensed)

- Added a subscription retry to the iPhone app (subscription can fail while pairing completes).

Files: CP3/iOS/PARMCO/PARMCO/PARMCOLink.swift

24 Nov 11, 3:48 AM

Prompt (team, verbatim)

> the pi is still waiting for the pop up to be accepted on the screen

Response (Claude, condensed)

- Advised accepting the pop-up once, and switching the Pi to console autologin before the final.
- Added a 3-second periodic re-claim of the default agent.

Files: CP3/Code/ble_server.c

25 Nov 11, 3:50 AM

Prompt (team, verbatim)

> lets try reverting to the origional code that had worked before discovering the bug

Response (Claude, condensed)

- Reverted ble_server.c to the pairing-agent version without the re-claim additions (22,699 bytes); kept the motor-thread and app changes.

Files: CP3/Code/ble_server.c

26 Nov 11, 3:56 AM

Prompt (team, verbatim)

> with this code, if i pair it for the first time, it works. however forgetting and repairing the device causes the bluetooth connection bug

Response (Claude, condensed)

- Diagnosed a stale bond on the Pi; instructed adding JustWorksRepairing = always to /etc/bluetooth/main.conf, restarting bluetooth, and removing the old bond.

Files: (Pi configuration, done by the team)

27 Nov 11, 4:16 AM

Prompt (team, verbatim)

> i think the bluetooth issue is resolved now. however we now have a problem with angle targetting.
>
> <- CALIBRATE_ZERO / -> ACK:CALIBRATE_ZERO / <- ANGLE:90.0 / -> ACK:ANGLE:90.00 / EVT:MOVE_DONE:target=90.00;angle=207.69;err=-117.69;ms=4072 / <- ANGLE:90.0 … EVT:MOVE_DONE:target=90.00;angle=158.64;err=-68.64;ms=4716 / … angle=81.47;err=8.53 / … angle=306.30;err=143.70 / <- SET_ZERO / EVT:MOVE_DONE:target=0.00;angle=163.48;err=-163.48;ms=2630

Response (Claude, condensed)

- Diagnosed a wrong step-to-encoder direction guess making corrections diverge.
- Added a 10-step settled direction probe, self-correction when a move goes the wrong way, MOVE_FAILED reporting and per-iteration diagnostics; tested normal, wrong-direction and stalled cases in simulation.

Files: CP3/Code/parmco_cp3_controller.c (4:18 version)

28 Nov 11, 4:20 AM

Prompt (team, verbatim)

> was any of the code changed by you in the past couple of minutes? it was working fine prior to the bluetooth issue

Response (Claude, condensed)

- Gave a timeline of every code change; identified the 3:05 motor-timing change as the only motor-related change before the problem.

Files: —

29 Nov 11, 4:22 AM

Prompt (team, verbatim)

> lets go back to the 2:30 version please

Response (Claude, condensed)

- Restored the controller byte-for-byte to the 2:30 version (45,407 bytes); kept the BLE pairing agent.

Files: CP3/Code/parmco_cp3_controller.c

30 Nov 11, 4:26 AM

Prompt (team, verbatim)

> accuracy is better, but we are about 3-5 degrees off at 180 and about 3 degrees off at 90

Response (Claude, condensed)

- Explained off-centre magnet error pattern; gave a hand-turn test and mechanical fixes (magnet centring and gap, protractor centring, parallax); offered a multi-point correction.

Files: —

31 Nov 11, 4:28 AM

Prompt (team, verbatim)

> never mind were good now

Response (Claude, condensed)

- Recommended committing and tagging; listed remaining CP3 deliverables.

Files: —

32 Nov 11, 4:33 AM

Prompt (team, verbatim)

> the motor spins ccw when it should be the other way around and vice versa

Response (Claude, condensed)

- Found the Mac copy had changed; asked which version was on the Pi (answer: the 4:18 version).
- Added #define CW_ASM_DIR 0 so CW maps to the correct assembly direction.

Files: CP3/Code/parmco_cp3_controller.c

33 Nov 11, 4:39 AM

Prompt (team, verbatim)

> the buttons for cw and ccw are still backwards

Response (Claude, condensed)

- Found the Mac file reverted to the 2:30 version; rewrote the 4:18 + direction-fix version and gave a commit-before-pull git sequence with verification greps.

Files: CP3/Code/parmco_cp3_controller.c

34 Nov 11, 4:54 AM

Prompt (team, verbatim)

> everything is working almost perfect. there is one final issue with the encoder check test B. encoder reading is greatly off from actual protractor reading.

Response (Claude, condensed)

- Gave a diagnosis table (mirrored direction, mechanical shift, coasting) with quick checks.

Files: —

35 Nov 11, 4:58 AM

Prompt (team, verbatim)

> the app is showing the position is 90 deg short of actual protractor reading. the blade was set to 0 of the protractor but the app reads 353.8

Response (Claude, condensed)

- Diagnosed slip between propeller hub or magnet and shaft during stalls; Sharpie-mark check; fixes (set screw on shaft flat, thread-locker, re-glue magnet).

Files: —

36 Nov 11, 5:01 AM

Prompt (team, verbatim)

> i retested and may have been user error. protractor is much closer this time around

Response (Claude, condensed)

- Recommended three repeat Test B runs, keeping Sharpie marks, per-session calibration check, committing and tagging.

Files: —

37 Nov 11, 7:24 AM

Prompt (team, verbatim)

> the rubric for checkpoint 3 is describing a GitHub site. are there any more details on the documentation about what the site entails?

Response (Claude, condensed)

- Quoted every site requirement from the project document (goal, authorship, rubric rows for CP3 and final) and suggested sections.

Files: —

38 Nov 11, 7:48 AM

Prompt (team, verbatim)

> could you create an skeleton for the github webpage. The flow of the page should be the title, followed by a an overview section. After the overview, the next section should have enough space to highlight the design targets we need to meet. Following the design targets, a short section with an extremly breif overview of each checkpoint that highlights the goals and timeline for the project.

Response (Claude, condensed)

- Built a GitHub Pages site (title with stepping protractor dial, overview, design-target spec sheet, checkpoint timeline) and checked it at desktop and mobile widths.

Files: finalProj/docs/index.html, styles.css

39 Nov 11, 8:01 AM

Prompt (team, verbatim)

> i only need the skeleton of the page and placeholders for material

Response (Claude, condensed)

- Stripped content to placeholders while keeping structure and styling.

Files: finalProj/docs/index.html, styles.css

40 Nov 11, 8:03 AM

Prompt (team, verbatim)

> keep a basic version of the hero image. that is a good idea

Response (Claude, condensed)

- Restored the protractor dial hero.

Files: finalProj/docs/index.html, styles.css

41 Nov 11, 11:34 AM

Prompt (team, verbatim)

> we are going to be completly scrapping and rebuilding the webpage. I am also unsure if you (AI) is supposed to be the entire author to the scripts. to be safe, you will be the main author of the code. lets start with the theme. I would like the website to be majority neutral colored with the University of GA colored accents. I do like the idea of the spinning motor, however, we can simplify this be removing the 90 deg subtext and all of the minor ticks. keep 0, 90, and the 180 deg ticks and the dual colored blades. As you scroll to the bottom of the website, the propeller spins from clockwise from 0, at the beginning of the page to 360 at the botton of the webpage. Additionally, the title should be fairly large when first opened and at the very top. When you begin to scroll, the title (left and center sections) should scale down and remain visible at the top. the spinning motor will do so similarly, but on the top right corner of the page.

Response (Claude, condensed)

- Confirmed UGA brand colours and fonts (Bulldog Red #BA0C2F, Arch Black, Stegeman; Oswald, Merriweather Sans).
- Rebuilt the site: neutral palette with UGA accents, simplified red/black propeller dial, scroll-driven rotation 0°→360° in 1.8° steps, large title that condenses into a fixed bar with the dial shrinking to the top-right; tested scroll states.

Files: finalProj/docs/index.html, styles.css, script.js

42 Nov 11, 11:39 AM

Prompt (team, verbatim)

> i did forget to something. I also need a fourth tick at 270 deg on the motor dial please

Response (Claude, condensed)

- Added the 270° tick and label.

Files: finalProj/docs/index.html

43 Nov 11, 11:44 AM

Prompt (team, verbatim)

> what is the default domain?

Response (Claude, condensed)

- Explained https://<owner>.github.io/<repo>/, private-repo and Enterprise caveats.

Files: —

44 Nov 11, 11:49 AM

Prompt (team, verbatim)

> the build errored out. below is the error message:
>
> Error: Logging at level: debug GitHub Pages: github-pages v232 GitHub Pages: jekyll v3.10.0 Theme: jekyll-theme-primer … Source: /github/workspace/./docs Destination: /github/workspace/./docs/_site … Rendering: assets/css/style.scss … github-pages 232 | Error: No such file or directory @ dir_chdir0 - /github/workspace/docs

Response (Claude, condensed)

- Diagnosed docs/ missing from the published branch root; added .nojekyll; gave commit/push and Pages-setting checks.

Files: finalProj/docs/.nojekyll

45 Nov 11, 11:56 AM

Prompt (team, verbatim)

> got it working. /doc was not under /root of the repo. could you update the motor tick marks again, the 270 changes were not staged/.

Response (Claude, condensed)

- Requested access to Group-2/docs (repo root) and wrote the 270° tick there.

Files: Group-2/docs/index.html

46 Nov 11, 12:02 PM

Prompt (team, verbatim)

> lets take a couple steps in a different direction. we have wiring schematic for the pi, octocoupler, motor, and hbridge. we need to add the magnetic motor encoder to the schematic. follow the same color coding rules we used for the schematic. let me know if you need me to list them back out

Response (Claude, condensed)

- Added U5 AS5600 (VCC 3V3 pin 17 green, GND pin 9 black, SDA/SCL GPIO2/3 yellow, DIR to GND black, OUT/GPO NC), magnet note and Rev C notes.

Files: CP3/Documentation/parmco_motor_schematic.png / .svg (Rev C)

47 Nov 11, 12:05 PM

Prompt (team, verbatim)

> can you connect all ground nodes so that they are common ground?

Response (Claude, condensed)

- Replaced separate ground symbols with one drawn common-ground network (left/middle collectors, top rail, bottom bus, single ground symbol); fixed overlaps.

Files: CP3/Documentation/parmco_motor_schematic.png / .svg

48 Nov 11, 12:16 PM

Prompt (team, verbatim)

> Go ahead and add a placeholder for an image of our physical project right below the overview header. Format, under the header space for me to type the overview of the project on the left half of the screen. To the right of the overview placeholder, I need an image placeholder to insert a high level flow diagram of the project's intended app and RP4 logic.

Response (Claude, condensed)

- Added photo placeholder and a text/flow-diagram two-column layout; noted the authorship rule for human-written site text.

Files: Group-2/docs/index.html, styles.css

49 Nov 11, 12:24 PM

Prompt (team, verbatim)

> could you update the files again? the changes are not populating on the page

Response (Claude, condensed)

- Rewrote the files (they had reverted) and gave commit/push and cache-refresh steps.

Files: Group-2/docs/index.html, styles.css

50 Nov 11, 1:12 PM

Prompt (team, verbatim)

> made some edits to the page to fill in the placeholders. formatting scaling is off. additionally, i want to shift the high level logic flow to be shifted down to sit next to the second paragraoph. next to the first paragraph, i want to add another picture that shows the app.

Response (Claude, condensed)

- Observed team's human edits (course line, title, authors, two overview paragraphs, marked).
- Fixed the aspect-ratio stretch bug; restructured into two rows: paragraph 1 + app image, paragraph 2 + flow diagram.

Files: Group-2/docs/index.html, styles.css

51 Nov 11, 1:16 PM

Prompt (team, verbatim)

> i think you removed my additions

Response (Claude, condensed)

- Found Claude's write had sent a stale cached copy that overwrote the team's edits; restored the correct version with fresh file names and verified it byte-for-byte.

Files: Group-2/docs/index.html, styles.css

52 Nov 11, 1:19 PM

Prompt (team, verbatim)

> can you change the way links are highlighted? the red underscore make it look like a spelling mistake underline. could you bold the links instead in gray?

Response (Claude, condensed)

- Links restyled bold grey without underline.

Files: Group-2/docs/styles.css

53 Nov 11, 1:27–1:29 PM

Prompt (team, verbatim)

> how would i view the screen layout simulation on xcode for the app? / yes please / and before i push, could you also change the links from gray to the uga red?

Response (Claude, condensed)

- Explained Canvas preview and Simulator, and that controls only appear after connecting.
- Added demo mode (sample readings, local button handling) used automatically in the Simulator and a #Preview block; real iPhone unaffected.
- Changed links to bold Bulldog Red.

Files: CP3/Code/iOS/PARMCO/PARMCO/PARMCOLink.swift, PARMCOApp.swift, ContentView.swift; Group-2/docs/styles.css

54 Nov 11, 1:40 PM

Prompt (team, verbatim)

> i accidentally overwrote the link color with a previous wepage version. can you readd those changes

Response (Claude, condensed)

- Verified the red-link stylesheet was still on disk; explained possible causes (not pushed, older push, unsaved editor tab, cache).

Files: —

55 Nov 11, 3:26 PM

Prompt (team, verbatim)

> can you fix the footers for me please

Response (Claude, condensed)

- Rebuilt footer as labelled columns (Team, Repository, AI tools) with a working repository link and a base line; kept team text; flagged TensorFlow mention for accuracy.

Files: Group-2/docs/index.html, styles.css

56 Nov 11, 3:31 PM

Prompt (team, verbatim)

> i have pushed the changes but git is not deploying the updates for me to view the website

Response (Claude, condensed)

- Checked the repository's Actions page in the built-in browser: runs #16–#17 queued, #4–#15 succeeded; GitHub status operational; advised waiting, cancel/re-run, or contacting the instructor.

Files: —

57 Nov 11, 3:37 PM

Prompt (team, verbatim)

> can you export all of our chat history as a .doc and .pdf file, as per the project requirements?

Response (Claude, condensed)

- Produced this log as Word and PDF.

Files: PARMCO_AI_Log_Claude_2026-11-11.docx / .pdf

Key findings from the session

- Hardware faults found by inspection, not code: miswired coil pairs (rebuild), an unclamped motor stand, and a possibly slipping hub — each was found by checking against the protractor or wiring, which is the independent-reference practice the project asks for.
- Bluetooth on iOS requires BLE; on Raspberry Pi OS the desktop's Bluetooth applet competes with a program's pairing agent, and re-pairing after 'Forget' needs JustWorksRepairing = always in BlueZ.
- Version control: several changes were lost because files were overwritten before being committed, and once because a stale copy was written. Committing immediately after each change, and verifying files after writing, prevented further losses.